

# Omer Yasir Onal

Back-End & Applied AI Engineer

engomeryasironal@gmail.com · +90 532 309 0261 · Istanbul, Turkiye · omeryasironal.com

GitHub OmerYasirOnal · LinkedIn omeryasironal · Medium @engomeryasironal · HackerRank engomeryasironal

Open to remote / international roles (UTC+3)

## SUMMARY

Computer Engineering new-grad and back-end & applied-AI engineer - AI Fellow (1,500 of 31,700, ~top 5%) and IEEE SIU 2025 author. I build reliable, well-tested systems: at AKIS - a live multi-agent AI platform (akisflow.com) - I designed a workflow where nothing ships without human review and a real passing test, and I shipped UniSum to the App Store on my own Node/MySQL API. Hands-on with Node.js/Python back-ends, PostgreSQL, Docker and CI/CD, and LLM/agent orchestration. Open to junior / entry-level full-time (and freelance) roles.

## SKILLS

**Security & Trust:** Signed build verification, Human approval workflows, OAuth 2.1, Privacy & consent compliance, Rate-limit & abuse hardening

**Backend & APIs:** Node.js, Express, TypeScript, FastAPI (Python), PostgreSQL, MySQL/Sequelize, SQLite, REST, JWT & bcrypt, WebSocket

**AI & LLM Engineering:** multi-agent orchestration, RAG (pgvector, ChromaDB), Model Context Protocol (MCP), Anthropic/OpenAI/Gemini/OpenRouter APIs, PyTorch (GPT/transformer from scratch), MLX LoRA fine-tuning, transfer learning (TensorFlow/Keras)

**DevOps & Infra:** Docker, GitHub Actions CI/CD, Vercel, OCI self-hosting, pnpm monorepo, SwiftPM

**Frontend & Mobile:** React, Next.js, Astro, Tailwind, Three.js / R3F, Vite · Swift 6 / SwiftUI, Flutter / Dart, Capacitor, Firebase, App Store delivery

## EXPERIENCE

**AI Fellow** - Yapay Zeka ve Teknoloji Akademisi

Istanbul · Dec 2025 - Present

- Selected among 1,500 of 31,700 applicants for the fellowship.
- Programme run by Google Turkiye, Girişimcilik Vakfı and T3 with the Ministry of Industry & Technology.
- Intensive curriculum in generative AI, ML/DL fundamentals, and end-to-end project development.

**Software Engineer Intern** - Exedra HRTech

Istanbul · Jul 2025 - Aug 2025

- Built web product features and tests in a production HRTech environment.
- Worked in an Agile team with code review.

**Instructor / Mentor** - T3 Foundation - Deneyap Technology Workshops

Istanbul · 2024 - 2025

- Taught and mentored students in the Deneyap technology workshops.

**Freelance Software Developer** - Freelance

Istanbul · May 2024 - Jan 2025

- Delivered end-to-end client sites incl. A2 Reklam - a 341-page trilingual (TR/EN/AR) Astro build.
- Owned scope, timeline and SEO / Google Ads.

**Researcher (lead contributor)** - TUBITAK 2209-A

Istanbul · Jun 2023 - Jun 2024

- Applied ML (XGBoost / Random Forest) to sensor data for structural / concrete durability.
- Work produced a paper accepted at IEEE SIU 2025.

## PROJECTS

## AKIS

[akisflow.com](https://akisflow.com)

Live, open-source (Apache-2.0) multi-agent AI build platform: an orchestrator coordinating 5 specialized agents, with mandatory human review before anything ships and a second AI reviewer that catches issues first. A task only counts as done once a real, working test passes. ~1,800 backend + ~700 frontend automated tests passing; Docker image trimmed by 198MB. Self-hosted at [akisflow.com](https://akisflow.com).

TypeScript · Node.js · React · PostgreSQL/pgvector · Docker · Multi-LLM · MCP

## UniSum

[apps.apple.com/tr/app/unisum/id6742401580](https://apps.apple.com/tr/app/unisum/id6742401580)

Shipped to the App Store - native SwiftUI client backed by a self-built Node/Express/MySQL REST API with secure login, hardened against common web attacks.

Swift · SwiftUI · Node.js · Express · MySQL · REST

## newBrain

[github.com/OmerYasirOnal/newbrain](https://github.com/OmerYasirOnal/newbrain)

A ~42M-parameter causal GPT built from scratch in PyTorch (own BPE tokenizer, RoPE/SwiGLU) with MLX LoRA fine-tuning - the fundamentals beneath the frameworks.

Python · PyTorch · MLX · FastAPI · Three.js · ChromaDB

## A2 Reklam

[a2reklam.com](https://a2reklam.com)

Freelance corporate site live in production - 341-page trilingual (TR/EN/AR) Astro build with Schema.org local-SEO and a hardened contact form.

Astro · Tailwind CSS · PHP · Schema.org · GTM/GA4

## Studio

[github.com/OmerYasirOnal/studio](https://github.com/OmerYasirOnal/studio)

Multi-agent game-development framework (16 specialist agents coordinating via the filesystem); Three.js + Capacitor, zero paid APIs.

TypeScript · Three.js · React Three Fiber · Capacitor · FastAPI

[More on GitHub](#) → [github.com/OmerYasirOnal](https://github.com/OmerYasirOnal)

## PUBLICATIONS

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**Machine learning for structural / concrete durability prediction (XGBoost / Random Forest)** - IEEE SIU 2025, 2025

## WRITING

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Author of 11 Medium articles on AI agents - [medium.com/@engomeryasironal](https://medium.com/@engomeryasironal)

## CERTIFICATIONS

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- **Foundations of Project Management** - Google (Coursera), Jan 2026
- **Supervised Machine Learning: Regression and Classification** - DeepLearning.AI (Coursera), Nov 2025
- **Exploratory Data Analysis for Machine Learning** - Coursera, Aug 2023

## EDUCATION

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B.Sc. Computer Engineering (50% English-medium) - Fatih Sultan Mehmet Vakif University - Graduated June 2026

## LANGUAGES

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Turkish (Native) · English (B2)

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